

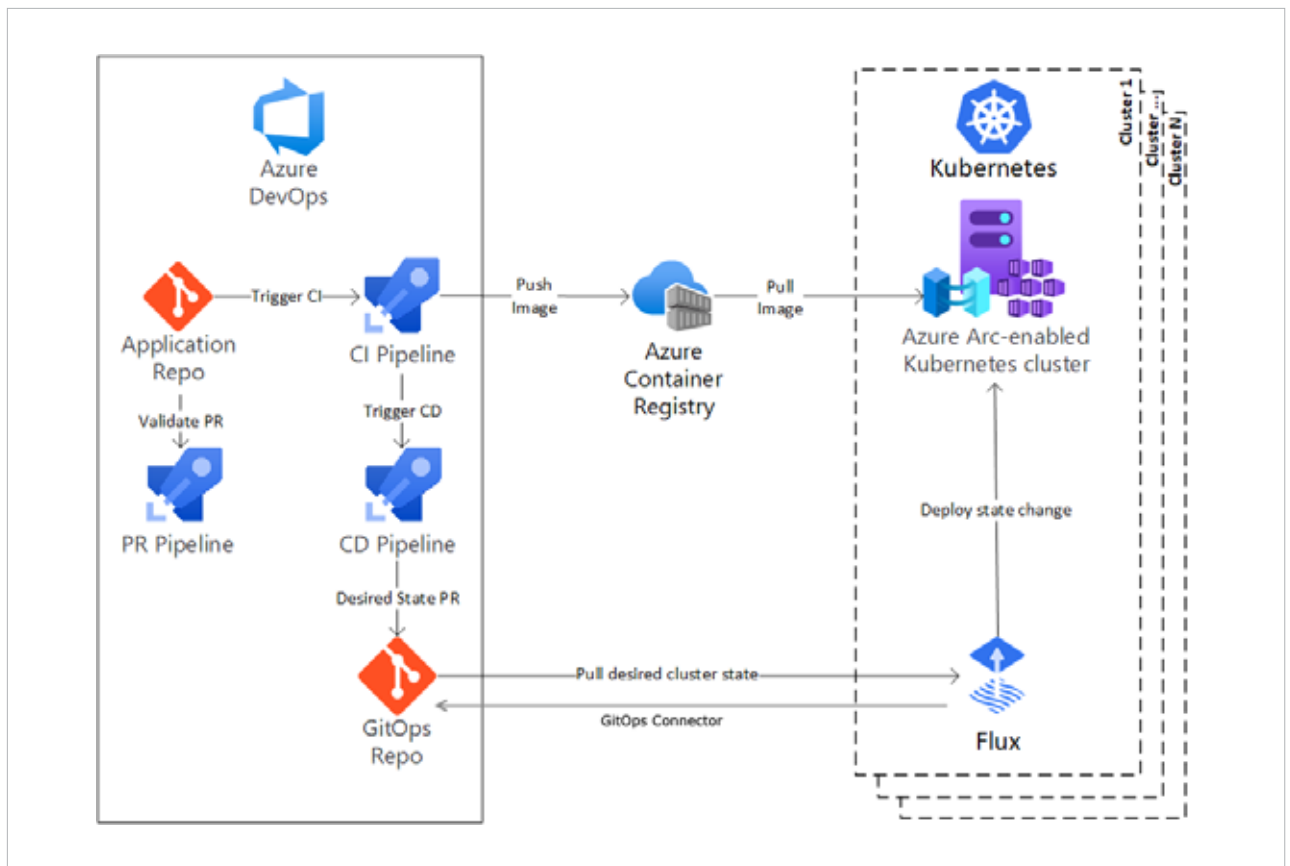


Figure 3: Flux

Flux

Flux is an open source continuous delivery tool that follows the GitOps approach to automate the deployment and management of applications in Kubernetes. Similar to Argo CD, Flux uses Git as the single source of truth for defining the desired state of applications and infrastructure configurations. It continuously monitors Git repositories and automatically synchronises the Kubernetes cluster with the configurations defined in those repositories.

How Flux works

Application definitions: You define the desired state of your applications and infrastructure configurations in a Git repository using Kubernetes manifests or Helm charts.

Flux controller: The Flux controller continuously monitors the

Git repository for changes. It checks for new commits and updates to the defined configurations.

Reconciliation: When changes are detected, the Flux controller initiates a reconciliation process. It compares the cluster's actual state with the desired state defined in the Git repository.

Automated deployment: Flux automatically applies the changes to the cluster, deploying and managing applications based on the configurations defined in Git.

Synchronisation: Flux keeps the cluster continuously synchronised with the Git repository. Any changes made directly to the cluster are reverted during the next synchronisation to ensure that the cluster remains in sync with the desired state.

Both Argo CD and Flux provide support for Helm charts, allowing you to deploy Helm releases using the CI/

CD pipeline. Additionally, you can set up custom policies in both tools to define the automated deployment behaviour, such as automated vs manual deployments and strategies for updating application versions.

It's essential to secure access to the Kubernetes cluster and the Git repository to prevent unauthorised changes. By integrating Argo CD or Flux with your CI/CD pipeline, you can establish a robust and automated GitOps-based workflow for deploying applications and configurations to your Kubernetes cluster with ease. **END** 🐧

By: Renganathan D.

The author is a Google cloud certified architect at HCL Technologies Ltd.

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